

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

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PRACTICAL 8

AIM :- Ethics and Bias Analysis

- A) Select any publicly available dataset (for example, Iris, Adult Income, or a classification dataset of your choice) and analyze it for possible bias and class imbalance. (Titanic Train Dataset)

CODE :-

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.read_csv("train.csv")

print("===== 1. LOAD AND EXPLORE DATASET =====")

print(df.head())

print("\nShape:", df.shape)

print("\nColumns:", df.columns.tolist())

print("\nDataset Information:")

df.info()

print("\nStatistical Summary:")

print(df.describe())

print("\n===== 2. CHECK DATA QUALITY =====")

print("\nMissing Values:")

print(df.isnull().sum())

print("\nDuplicate Rows:", df.duplicated().sum())

# 3. IDENTIFY TARGET VARIABLE

print("\n===== 3. TARGET VARIABLE =====")
```

```

target = "Survived"
print("Target Variable:", target)
print("\nTarget Values:")
print(df[target].value_counts())
print("\n===== 4. CLASS DISTRIBUTION =====")
print(df["Survived"].value_counts())
print("\nPercentage Distribution:")
print(df["Survived"].value_counts(normalize=True) * 100)
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Survived")
plt.title("Survival Class Distribution")
plt.xlabel("Survival")
plt.ylabel("Number of Passengers")
plt.xticks([0, 1], ["Did Not Survive", "Survived"])
plt.show()
print("\n===== 6. GENDER REPRESENTATION =====")
print(df["Sex"].value_counts())
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Sex")
plt.title("Gender Representation")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.show()
print("\n===== 7. SURVIVAL BY GENDER =====")
print(pd.crosstab(df["Sex"], df["Survived"]))
print("\nSurvival Rate by Gender:")
print(df.groupby("Sex")["Survived"].mean() * 100)
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Sex", hue="Survived")

```

```

plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.show()

print("\n===== 8. PASSENGER CLASS REPRESENTATION
=====")

print(df["Pclass"].value_counts().sort_index())

plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Pclass")
plt.title("Passenger Class Representation")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.show()

print("\n===== 9. SURVIVAL BY PASSENGER CLASS =====")
print(pd.crosstab(df["Pclass"], df["Survived"]))
print("\nSurvival Rate by Passenger Class:")
print(df.groupby("Pclass")["Survived"].mean() * 100)

plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Pclass", hue="Survived")
plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.show()

print("\n===== 10. AGE GROUPS =====")
bins = [0, 12, 18, 35, 60, 100]
labels = ["Child", "Teen", "Adult", "Middle-aged", "Senior"]
df["AgeGroup"] = pd.cut(
    df["Age"],
    bins=bins,
    labels=labels
)

```

```

print(df["AgeGroup"].value_counts().sort_index())
plt.figure(figsize=(7, 4))
sns.countplot(data=df, x="AgeGroup")
plt.title("Age Group Distribution")
plt.xlabel("Age Group")
plt.ylabel("Number of Passengers")
plt.xticks(rotation=20)
plt.show()

print("\n===== 11. SURVIVAL BY AGE GROUP =====")
print(pd.crosstab(df["AgeGroup"], df["Survived"]))
print("\nSurvival Rate by Age Group:")
print(df.groupby("AgeGroup", observed=True)["Survived"].mean() * 100)
plt.figure(figsize=(7, 4))
sns.countplot(data=df, x="AgeGroup", hue="Survived")
plt.title("Survival by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Number of Passengers")
plt.xticks(rotation=20)
plt.show()

print("\n===== 12. POSSIBLE BIAS =====")
print("Gender Survival Rate:")
print(df.groupby("Sex")["Survived"].mean() * 100)
print("\nPassenger Class Survival Rate:")
print(df.groupby("Pclass")["Survived"].mean() * 100)
print("\nPossible Bias:")
print("- Gender differences in survival")
print("- Passenger class differences")
print("- Age differences")
print("- Missing values")
print("- Unequal representation of groups")

```

```
print("\n===== 13. IMPACT ON MACHINE LEARNING =====")
```

```
print("""
```

1. Class imbalance may affect model performance.
2. Missing values can reduce prediction accuracy.
3. Gender and passenger class may strongly influence predictions.
4. The model may learn historical patterns and biases.
5. Accuracy alone may not be sufficient for evaluation.

```
""")
```

```
print("\n===== 14. SUGGESTED SOLUTIONS =====")
```

```
print("""
```

1. Handle missing Age values.
2. Handle missing Embarked values.
3. Drop or transform Cabin because of many missing values.
4. Use stratified train-test splitting.
5. Use class weights if required.
6. Evaluate using Precision, Recall and F1-score.
7. Check model performance across different groups.

```
""")
```

```
print("\n===== 15. FINAL CONCLUSION =====")
```

```
print("""
```

The Titanic dataset shows that survival was strongly related to gender, passenger class and age. Female and first-class passengers had higher survival rates, while male and third-class passengers had lower survival rates.

The dataset also contains missing values and differences in group representation. Therefore, proper data preprocessing, bias analysis and suitable evaluation metrics are important before applying machine learning.

```
""")
```

OUTPUT :-

```
*** ===== 1. LOAD AND EXPLORE DATASET =====
PassengerId  Survived  Pclass  \
0            1         0        3
1            2         1        1
2            3         1        3
3            4         1        1
4            5         0        3

                                Name    Sex  Age  SibSp  \
0                                Braund, Mr. Owen Harris  male  22.0    1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0    1
2                                Heikkinen, Miss. Laina  female  26.0    0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0    1
4                                Allen, Mr. William Henry  male  35.0    0

Parch    Ticket    Fare  Cabin  Embarked
0      0      A/5 21171   7.2500   NaN      S
1      0      PC 17599  71.2833   C85      C
2      0  STON/O2. 3101282   7.9250   NaN      S
3      0     113803  53.1000  C123      S
4      0    373450   8.0500   NaN      S

Shape: (891, 12)

Columns: ['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked']

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
Statistical Summary:
PassengerId  Survived  Pclass    Age    SibSp
count  891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std    257.353842    0.486592    0.836071   14.526497    1.102743
min      1.000000    0.000000    1.000000    0.420000    0.000000
25%    223.500000    0.000000    2.000000   20.125000    0.000000
50%    446.000000    0.000000    3.000000   28.000000    0.000000
75%    668.500000    1.000000    3.000000   38.000000    1.000000
max    891.000000    1.000000    3.000000   80.000000    8.000000

Parch    Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std     0.806057   49.693429
min     0.000000    0.000000
25%     0.000000    7.910400
50%     0.000000   14.454200
75%     0.000000   31.000000
max     6.000000  512.329200
```

```
===== 2. CHECK DATA QUALITY =====

Missing Values:
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age           177
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin         687
Embarked       2
dtype: int64

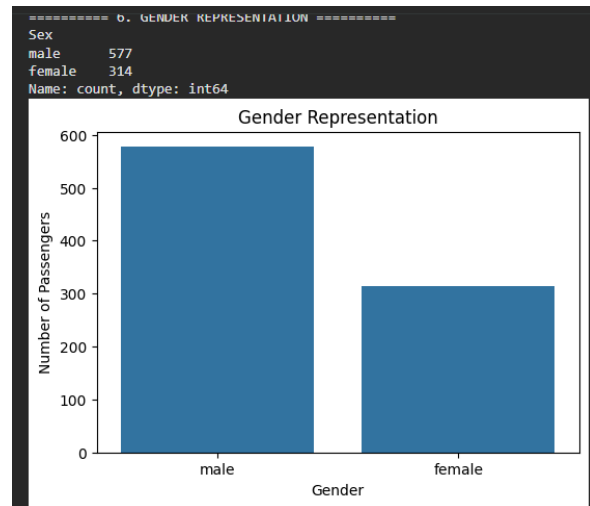
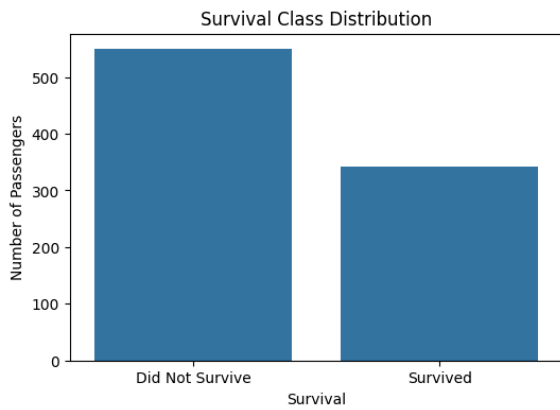
Duplicate Rows: 0
```

```
===== 3. TARGET VARIABLE =====
Target Variable: Survived

Target Values:
Survived
0      549
1      342
Name: count, dtype: int64

===== 4. CLASS DISTRIBUTION =====
Survived
0      549
1      342
Name: count, dtype: int64

Percentage Distribution:
Survived
0      61.616162
1      38.383838
Name: proportion, dtype: float64
```

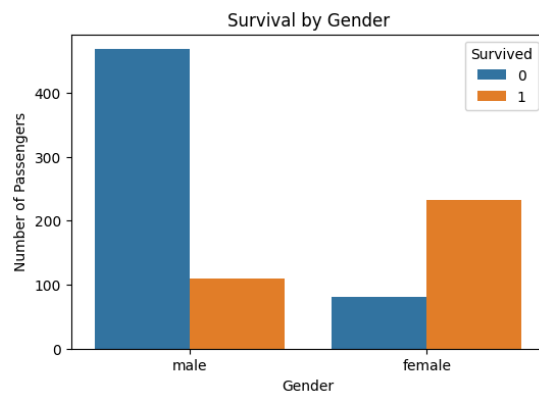


```

===== 7. SURVIVAL BY GENDER =====
Survived  0    1
Sex
female    81   233
male     468   109

Survival Rate by Gender:
Sex
female    74.203822
male     18.890815
Name: Survived, dtype: float64

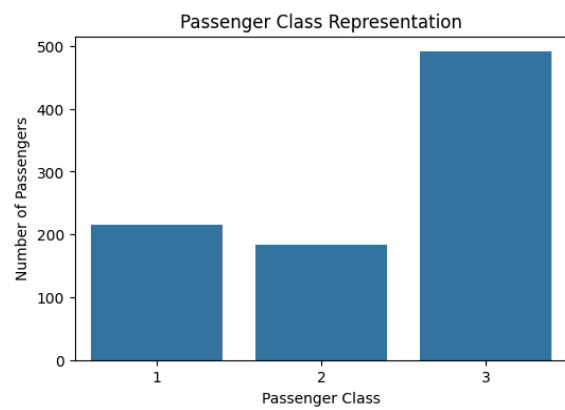
```



```

===== 8. PASSENGER CLASS REPRESENTATION =====
Pclass
1      216
2      184
3      491
Name: count, dtype: int64

```

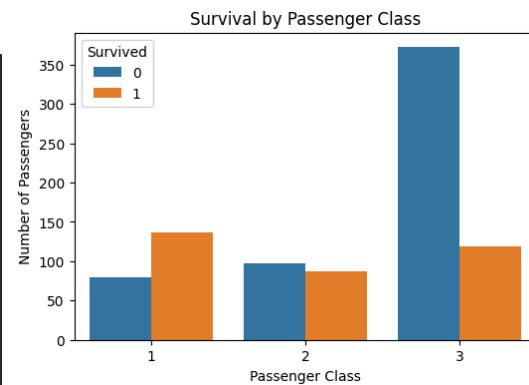


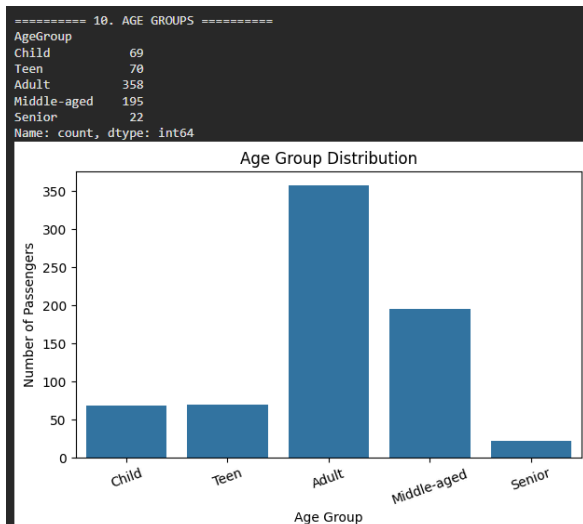
```

===== 9. SURVIVAL BY PASSENGER CLASS =====
Survived  0    1
Pclass
1         80   136
2         97    87
3        372   119

Survival Rate by Passenger Class:
Pclass
1     62.962963
2     47.282609
3     24.236253
Name: Survived, dtype: float64

```



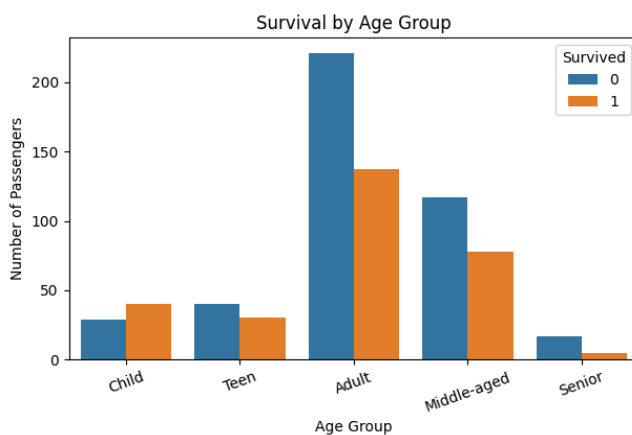


```

===== 11. SURVIVAL BY AGE GROUP =====
Survived    0    1
AgeGroup
Child        29   40
Teen         40   30
Adult        221  137
Middle-aged  117   78
Senior        17    5

Survival Rate by Age Group:
AgeGroup
Child      57.971014
Teen       42.857143
Adult      38.268156
Middle-aged 40.000000
Senior     22.727273
Name: Survived, dtype: float64

```



```

===== 12. POSSIBLE BIAS =====
Gender Survival Rate:
Sex
female  74.203822
male    18.890815
Name: Survived, dtype: float64

Passenger Class Survival Rate:
Pclass
1      62.962963
2      47.282609
3      24.236253
Name: Survived, dtype: float64

Possible Bias:
- Gender differences in survival
- Passenger class differences
- Age differences
- Missing values
- Unequal representation of groups

```

IMPACT ON MACHINE LEARNING

1. Class imbalance may affect model performance.
2. Missing values can reduce prediction accuracy.
3. Gender and passenger class may strongly influence predictions.
4. The model may learn historical patterns and biases.
5. Accuracy alone may not be sufficient for evaluation.

SUGGESTED SOLUTIONS

1. Handle missing Age values.
2. Handle missing Embarked values.
3. Drop or transform Cabin because of many missing values.
4. Use stratified train-test splitting.
5. Use class weights if required.
6. Evaluate using Precision, Recall and F1-score.
7. Check model performance across different groups.

FINAL CONCLUSION

The Titanic dataset shows that survival was strongly related to gender, passenger class and age. Female and first-class passengers had higher survival rates, while male and third-class passengers had lower survival rates. The dataset also contains missing values and differences in group representation. Therefore, proper data preprocessing, bias analysis and suitable evaluation metrics are important before applying machine learning.

B) German Credit Risk Dataset

CODE :-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("german_credit_data.csv")
print("===== 1. LOAD AND EXPLORE DATASET =====")
print(df.head())
print("\nDataset Shape:")
print(df.shape)
print("\nColumns:")
print(df.columns.tolist())
print("\nDataset Information:")
df.info()
print("\nStatistical Summary:")
print(df.describe())
print("\n===== 2. CHECK DATA QUALITY =====")
print("\nMissing Values:")
print(df.isnull().sum())
print("\nDuplicate Rows:")
print(df.duplicated().sum())
print("\n===== 3. IDENTIFY TARGET VARIABLE =====")
possible_targets = ["Risk", "risk", "Credit Risk", "Class", "class"]
target = None
for col in possible_targets:
    if col in df.columns:
        target = col
        break
if target:
    print("Target Variable:", target)
else:
    print("Credit Risk target variable is NOT present in this dataset.")
    print("Available columns are:")
    print(df.columns.tolist())
print("\n===== 4. CLASS DISTRIBUTION =====")
if target:
    print(df[target].value_counts())
```

```

print("\nPercentage Distribution:")
    print(df[target].value_counts(normalize=True) * 100)
else:
    print("Cannot analyze credit-risk class distribution.")
    print("Reason: Target variable is missing.")
print("\n===== 5. CLASS IMBALANCE =====")
if target:
    plt.figure(figsize=(6, 4))
    sns.countplot(data=df, x=target)
    plt.title("Credit Risk Class Distribution")
    plt.xlabel("Credit Risk")
    plt.ylabel("Number of Customers")
    plt.show()
else:
    print("Cannot display credit-risk class imbalance.")
    print("Reason: Credit Risk target variable is missing.")
print("\n===== 6. GENDER REPRESENTATION =====")
print(df["Sex"].value_counts())
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Sex")
plt.title("Gender Representation")
plt.xlabel("Gender")
plt.ylabel("Number of Customers")
plt.show()
print("\n===== 7. CREDIT RISK BY GENDER =====")
if target:
    print(pd.crosstab(df["Sex"], df[target]))
    plt.figure(figsize=(6, 4))
    sns.countplot(data=df, x="Sex", hue=target)
    plt.title("Credit Risk by Gender")
    plt.xlabel("Gender")
    plt.ylabel("Number of Customers")
    plt.show()
else:
    print("Cannot analyze Credit Risk by Gender.")
    print("Reason: Credit Risk target variable is missing.")
print("\n===== 8. AGE GROUPS =====")
bins = [0, 18, 30, 45, 60, 100]
labels = [
    "18 or below",
    "19-30",
    "31-45",
    "46-60",
    "60+"

```

```

]
df["AgeGroup"] = pd.cut(
    df["Age"],
    bins=bins,
    labels=labels
)
print(df["AgeGroup"].value_counts().sort_index())
plt.figure(figsize=(7, 4))
sns.countplot(
    data=df,
    x="AgeGroup",
    order=labels
)
plt.title("Age Group Distribution")
plt.xlabel("Age Group")
plt.ylabel("Number of Customers")
plt.xticks(rotation=20)
plt.show()
print("\n===== 9. CREDIT RISK BY AGE GROUP =====")
if target:
    print(pd.crosstab(df["AgeGroup"], df[target]))
    plt.figure(figsize=(8, 4))
    sns.countplot(
        data=df,
        x="AgeGroup",
        hue=target,
        order=labels
    )
    plt.title("Credit Risk by Age Group")
    plt.xlabel("Age Group")
    plt.ylabel("Number of Customers")
    plt.xticks(rotation=20)
    plt.show()
else:
    print("Cannot analyze Credit Risk by Age Group.")
    print("Reason: Credit Risk target variable is missing.")
print("\n===== 10. HOUSING CATEGORY REPRESENTATION =====")
print(df["Housing"].value_counts())
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Housing")
plt.title("Housing Category Representation")
plt.xlabel("Housing Category")
plt.ylabel("Number of Customers")

```

```

plt.show()
print("\n===== 11. CREDIT RISK BY HOUSING CATEGORY
=====")
if target:
    print(pd.crosstab(df["Housing"], df[target]))
    plt.figure(figsize=(7, 4))
    sns.countplot(
        data=df,
        x="Housing",
        hue=target
    )
    plt.title("Credit Risk by Housing Category")
    plt.xlabel("Housing Category")
    plt.ylabel("Number of Customers")
    plt.show()
else:
    print("Cannot analyze Credit Risk by Housing Category.")
    print("Reason: Credit Risk target variable is missing.")
print("\n===== 12. POSSIBLE BIAS =====")
print("""
Possible sources of bias include:
1. Gender representation may be unequal.
2. Some age groups may have more customers than others.
3. Housing categories may not be equally represented.
4. Missing values may affect analysis.
5. If a Credit Risk target is added later, differences
   between demographic groups should be checked carefully.
""")

```

OUTPUT :-

```

===== 1. LOAD AND EXPLORE DATASET =====
Unnamed: 0  Age  Sex  Job  Housing  Saving accounts  Checking account  \
0          0   67  male   2    own             NaN             little
1          1   22  female  2    own             little          moderate
2          2   49  male   1    own             little             NaN
3          3   45  male   2    free             little             little
4          4   53  male   2    free             little             little

   Credit amount  Duration  Purpose
0          1169         6  radio/TV
1          5951        48  radio/TV
2          2096        12  education
3          7882        42  furniture/equipment
4          4870        24         car

Dataset Shape:
(1000, 10)

Columns:
['Unnamed: 0', 'Age', 'Sex', 'Job', 'Housing', 'Saving accounts', 'Checking account', 'Credit amount', 'Duration', 'Purpose']

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Unnamed: 0          1000 non-null   int64

```

```

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype  
---  -
0   Unnamed: 0          1000 non-null   int64  
1   Age                 1000 non-null   int64  
2   Sex                 1000 non-null   object  
3   Job                 1000 non-null   int64  
4   Housing             1000 non-null   object  
5   Saving accounts     817 non-null    object  
6   Checking account    606 non-null    object  
7   Credit amount       1000 non-null   int64  
8   Duration            1000 non-null   int64  
9   Purpose             1000 non-null   object  
dtypes: int64(5), object(5)
memory usage: 78.3+ KB

Statistical Summary:
      Unnamed: 0      Age      Job      Credit amount      Duration
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean    499.500000   35.546000   1.904000   3271.258000   20.903000
std    288.819436   11.375469   0.653614   2822.736876   12.058814
min      0.000000   19.000000   0.000000   250.000000   4.000000
25%    249.750000   27.000000   2.000000   1365.500000   12.000000
50%    499.500000   33.000000   2.000000   2319.500000   18.000000
75%    749.250000   42.000000   2.000000   3972.250000   24.000000
max    999.000000   75.000000   3.000000  10414.000000   71.000000

```

```

===== 2. CHECK DATA QUALITY =====

Missing Values:
Unnamed: 0      0
Age             0
Sex             0
Job             0
Housing         0
Saving accounts 183
Checking account 394
Credit amount   0
Duration         0
Purpose         0
dtype: int64

Duplicate Rows:
0

```

```

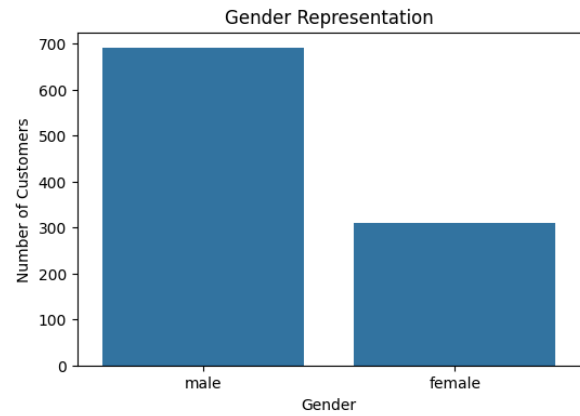
===== 3. IDENTIFY TARGET VARIABLE =====
Credit Risk target variable is NOT present in this dataset.
Available columns are:
['Unnamed: 0', 'Age', 'Sex', 'Job', 'Housing', 'Saving accounts', 'Checking account', 'Credit amount', 'Duration', 'Purpose']

===== 4. CLASS DISTRIBUTION =====
Cannot analyze credit-risk class distribution.
Reason: Target variable is missing.

===== 5. CLASS IMBALANCE =====
Cannot display credit-risk class imbalance.
Reason: Credit Risk target variable is missing.

===== 6. GENDER REPRESENTATION =====
Sex
male      690
female    310
Name: count, dtype: int64

```



```

===== 7. CREDIT RISK BY GENDER =====
Cannot analyze Credit Risk by Gender.
Reason: Credit Risk target variable is missing.

===== 8. AGE GROUPS =====
AgeGroup
18 or below      0
19-30            411
31-45            403
46-60            141
60+              45
Name: count, dtype: int64

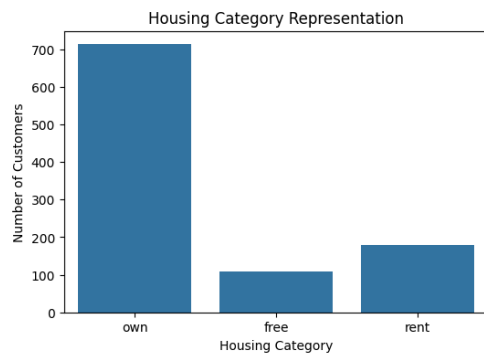
```

```

===== 9. CREDIT RISK BY AGE GROUP =====
Cannot analyze Credit Risk by Age Group.
Reason: Credit Risk target variable is missing.

===== 10. HOUSING CATEGORY REPRESENTATION =====
Housing
own      713
rent     179
free     108
Name: count, dtype: int64

```



```
===== 11. CREDIT RISK BY HOUSING CATEGORY =====
Cannot analyze Credit Risk by Housing Category.
Reason: Credit Risk target variable is missing.
```

```
===== 12. POSSIBLE BIAS =====
```

```
Possible sources of bias include:
```

1. Gender representation may be unequal.
2. Some age groups may have more customers than others.
3. Housing categories may not be equally represented.
4. Missing values may affect analysis.
5. If a Credit Risk target is added later, differences between demographic groups should be checked carefully.

13. Impact on ML

The absence of a **Credit Risk target variable** prevents supervised machine-learning models from being trained to predict credit risk. If a target column is added, unequal representation of gender, age groups, and housing categories could cause the model to perform differently across groups. Missing values and class imbalance could also reduce model performance.

14. Suggested Solutions

The dataset should first be completed by adding the correct **Credit Risk/Target** column. Missing values should then be handled appropriately. If the target classes are imbalanced, techniques such as class weighting or resampling can be used. Stratified train-test splitting and evaluation using precision, recall, F1-score, and confusion matrix are also recommended. Model performance should additionally be checked across gender, age, and housing groups to identify possible bias.

15. Final Conclusion

The German Credit dataset contains useful information about **age, gender, housing, savings, checking accounts, credit amount, duration, and purpose**. The analysis can identify representation patterns in gender, age, and housing. However, the uploaded dataset does not contain the **Credit Risk target variable**, so actual credit-risk analysis and supervised ML prediction cannot be performed until that column is provided.